

FINAL EXPLANATION: PHYSICS GRADE 10

Teacher Key: Exemplar Answers and Marking Guide

FINAL EXPLANATION: PHYSICS GRADE 10 – SUB-STRAND 1.2 MECHANICAL PROPERTIES OF MATERIALS: FINAL EXPLANATION

Teacher Key: Exemplar Answers (do not give to students)

INSTRUCTIONS GIVEN TO STUDENTS

Imagine you are advising the farmer in Kericho whose sisal footbridge ropes behaved so differently under an 80 kg sack of tea leaves. Answer each question in the space under it, in your own words, using evidence from Lessons 1 to 6, your Summary Table, your force-extension graphs and your Final Model. Use correct vocabulary (elastic, plastic, elastic limit, spring constant, Hooke's Law, elastic potential energy, stress, strain, Young's Modulus). Show every calculation step with units, and explain WHY and HOW, not just WHAT. Take $g = 9.8 \text{ N/kg}$. You may add labelled diagrams. You have 80 to 100 minutes and you work individually.

Part 1: Elastic and Plastic Behaviour – What Did We Observe?

Remember the two sisal ropes on the Kericho suspension footbridge, and the three materials you stretched in Lessons 1 and 2 (rubber band, sisal twine, copper wire). (a) Describe what happened to each rope when the 80 kg sack was loaded and then removed. (b) Describe what you observed for each of the three materials in your stretching investigation. (c) Define elastic behaviour and plastic deformation, and use your observations to classify each material. (d) Explain why two ropes that look similar can behave so differently.

(a) The new sisal rope (Rope A) stretched under the 80 kg sack and returned to its original length when the load was removed. The old sisal rope (Rope B) stretched, did not return, and then snapped. (b) The rubber band returned to its original length at every load from 100 g to 500 g. The sisal twine returned at low loads but showed permanent lengthening beyond about 300 g. The copper wire stayed bent even after a small load of 100–200 g. (c) Elastic behaviour means a material returns to its original shape and size after the force is removed. Plastic deformation means the material stays permanently changed in shape after the force is removed. The rubber band is elastic, the copper wire is plastic, and the sisal twine is elastic at low loads but becomes plastic once the load passes about 300 g. (d) The outside of a rope shows little about its inside. The old rope's fibres had been degraded by age, sun and repeated loading, so it could not recover from stretching as the new rope could. The ropes differ in a measurable internal property, not in appearance.

Part 2: What Is Happening Inside the Material?

Use a labelled particle-level diagram and a short explanation to show what

Diagram (described): Elastic case – particles are drawn evenly spaced. Under force the spaces between them become larger and the bonds are drawn stretched but still connected. When the force is removed the particles are drawn back in their original positions. Beyond the elastic limit – particles are drawn displaced into new positions, with some bonds broken or the particles

happens to the particles and the bonds between them (a) when an elastic material is stretched and the force is removed, and (b) when a material is stretched beyond its elastic limit. Then say how your thinking about this has changed since the first model you sketched in Lesson 1.

unable to return. (a) In an elastic material the force stretches the bonds between particles but does not break them. The bonds act like tiny springs that pull the particles back, so the material returns to its original shape. (b) Beyond the elastic limit the force moves particles further than their restoring bonds can pull them back. They settle in new positions, so the deformation is permanent. If the force keeps acting, the bonds break and the material fractures, as the old Kericho rope did. Growth: in Lesson 1 my model only showed particles moving apart, with no explanation of why some materials recover. Now my model includes restoring bonds, the elastic limit, and a link to measurable quantities such as the spring constant.

Part 3: Hooke's Law, Elastic Limit and Stored Energy – Calculating for the Kericho Ropes

Use this data about the two ropes, with the load being the 80 kg tea-leaf sack.

Rope	Spring constant k (N/m)	Extension under the 80 kg load (m)
Rope A (new)	850	0.92
Rope B (old)	210	3.73

The elastic limit of Rope B is an extension of 2.1 m. (a) Calculate the weight of the sack. (b) State Hooke's Law and explain what the spring constant tells you about a material. (c) Calculate the force at Rope B's elastic limit and the mass this represents (its safe working load). (d) Calculate the elastic potential energy stored in Rope B at its elastic limit, and in Rope A at an extension of 0.92

(a) $W = mg = 80 \times 9.8 = 784 \text{ N}$ (about 800 N). (b) Hooke's Law: within the elastic limit, the extension of a material is directly proportional to the applied force, $F = kx$. The spring constant k (N/m) is the gradient of the linear part of the force-extension graph. A larger k means a stiffer material that needs more force for each metre of extension. (c) $F = kx = 210 \times 2.1 = 441 \text{ N}$. Safe working load as a mass: $441 / 9.8 = 45 \text{ kg}$, far less than the 80 kg sack. (d) Rope B at its elastic limit: $E_p = \frac{1}{2}kx^2 = \frac{1}{2} \times 210 \times 2.1^2 = 105 \times 4.41 = 463 \text{ J}$ (about 463 J). Rope A at 0.92 m: $E_p = \frac{1}{2} \times 850 \times 0.92^2 = 425 \times 0.8464 = 360 \text{ J}$ (about 360 J). (e) The sack's weight of 784 N is much more than the 441 N that Rope B can take within its elastic limit. Its extension of 3.73 m is nearly double its elastic limit of 2.1 m, so it deformed plastically and then snapped. Rope A has a much larger k (850 N/m), so the same force stretches it only 0.92 m. That is within its elastic limit, so it recovered when the load was removed. Graph: Rope A is a steeper line, Rope B a shallower line that curves away from linear at 2.1 m. A vertical line at 784 N lies beyond Rope B's elastic limit but within Rope A's linear region.

m. (e) Using your answers, explain why Rope B failed and Rope A did not. Sketch the two force-extension graphs side by side and mark the load and the elastic limit of Rope B.	
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Part 4: Stress, Strain and Young's Modulus – Predicting Before Loading

(a) Define stress, strain and Young's Modulus, giving the equation and the unit (if any) for each. (b) Explain why a farmer or engineer should compare stress and strain rather than only force and extension when choosing a rope. (c) Name and describe the key features of a stress-strain graph that you identified in Lesson 5. (d) Explain how measuring Young's Modulus and fracture stress BEFORE loading would have let the farmer predict which rope would snap.	(a) Stress $\sigma = F/A$ is the force acting per unit cross-sectional area, measured in pascals (Pa). Strain $\epsilon = \Delta L/L_0$ is the extension compared with the original length. It has no unit. Young's Modulus $E = \sigma/\epsilon$ is a constant for a material that measures its stiffness. A higher E means more stress is needed to produce the same strain. (b) Force and extension depend on the size of the sample: a thicker or longer rope gives different values from a thin or short one made of the same fibre. Stress and strain take the cross-sectional area and the original length into account, so they describe the material itself and allow fair comparison of different ropes. (c) The graph has a linear (Hookean) region where stress is proportional to strain, the elastic limit beyond which the material does not fully recover, the yield point where the material starts to deform plastically, the ultimate tensile strength (UTS) which is the maximum stress, and the fracture point where it breaks. (d) The old rope's fibres were degraded by years of use, sun and repeated loading. This gave it a lower Young's Modulus and a fracture stress below the stress produced by the 80 kg sack. If the farmer compared each rope's measured E and fracture stress with the stress the load produces (F/A), he would see that the old rope would be overloaded while the new rope would stay in its elastic region. So the failure could have been predicted with numbers instead of guesswork.
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Part 5: Final Explanation – Advice to the Kericho Farmer

Write a formal recommendation to the Kericho farmer that answers the driving question: Why do some materials stretch and snap back while others permanently deform or break, and how can we use this knowledge to choose the right material for a safe structure in Kenya? Your answer must (a) explain why the two ropes behaved differently, using the elastic limit, spring constant and energy ideas; (b) give a safe working load for the old	Dear Farmer, the two ropes looked the same but behaved differently because of what happens inside them when they are stretched. Within its elastic limit a material obeys Hooke's Law ($F = kx$): the bonds between particles stretch and pull the particles back when the load is removed, and the material stores elastic potential energy ($E_p = \frac{1}{2}kx^2$). Your new rope has a high spring constant (850 N/m), so your 80 kg sack (about 784 N) stretches it only 0.92 m. This is within its elastic limit, so it recovered. Your old rope has fibres weakened by age, sun and repeated loading. Its spring constant is only 210 N/m and its elastic limit is an extension of 2.1 m. The same sack tried to stretch it 3.73 m, nearly double that limit. It deformed permanently, its bonds were pulled apart, and it snapped. Its safe working load is the force at its elastic limit, $210 \times 2.1 = 441$ N, or about 45 kg, well below your 80 kg sack. I recommend that you replace the old rope with one of high spring constant and high Young's Modulus, and never load a rope beyond its safe working load. Check that the stress from your load is below the fracture stress of the rope. The same reasoning applies elsewhere. A matatu suspension spring must have a suitable k so that bumps on Nairobi roads compress it within its elastic limit and it recovers each time. A bungee cord at a Nairobi adventure park must store energy as $E_p = \frac{1}{2}kx^2$ without passing its elastic limit. Growth: in Lesson 1 I only predicted that the old rope was 'weaker'. Now I can explain with particle bonds, graphs, calculations and stress and strain exactly why it failed, and I can predict before loading which rope is safe.
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rope and say what you recommend he does; (c) describe one other Kenyan application, such as a matatu suspension spring in Nairobi or a bungee cord at a Nairobi adventure park, where the same ideas help choose a safe material; and (d) explain how your understanding has grown from your first prediction in Lesson 1.	
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RUBRIC			
Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Understanding of mechanical properties	Defines elastic, plastic, elastic limit, spring constant, stress, strain and Young's Modulus accurately, and links them to a particle-level explanation of bonds.	Defines most terms correctly with a minor gap or imprecision in the particle-level explanation.	Definitions are incomplete or confused, for example mixing up elastic limit and breaking point, and there is little or no particle-level explanation.
Use of evidence	Uses specific data from the lessons accurately (the 100–500 g stretching results, k values of 850 and 210 N/m, extensions of 0.92 m and 3.73 m, elastic limit 2.1 m).	Uses some specific lesson evidence, with a few missing or inexact values.	Uses general statements with little or no specific data from the investigations.
Scientific reasoning and calculations	Calculations (weight, safe working load, E_p) are correct, with steps and units shown, and the reasoning links the load to the elastic limit to the failure.	Calculations are mostly correct with small errors in units or steps, and the reasoning is mostly complete.	Calculations have major errors or are missing, and the reasoning is a list of facts that does not explain why the rope snapped.
Application to Kenyan contexts	Gives a clear, practical recommendation to the farmer and explains a second Kenyan application (matatu suspension spring or bungee cord) correctly.	Gives a reasonable recommendation and mentions a second application with limited explanation.	Recommendation is vague or unsupported, and there is no clear second application.
Communication and growth	Writing is clear, organised and uses correct vocabulary and labelled diagrams, and shows a clear journey from the Lesson 1 prediction to evidence-based explanation.	Writing is mostly clear and organised, with some growth from Lesson 1 described.	Writing is hard to follow, with little vocabulary use and no description of how understanding changed.

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